

USER GUIDE

SMR Grid Integration Simulator

A Step-by-Step Guide for Any User

No engineering, energy, or programming background required

Companion to: [SMR_Grid_Integration_Simulator.html](#)

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Version 1.0 — First Version

September 2026

1. How to Use This Guide

This guide teaches you to use the SMR Grid Integration Simulator from zero. You do not need to know anything about power plants, nuclear energy, or spreadsheets. Every technical word is explained in Section 3, and every step tells you what you should see afterwards so you can check that you are on track.

1.1 Who It Is For

Students, teachers, journalists, policy staff, planners, and curious readers in any country. The tool is a general calculator: it does not contain data for any particular country until you type your own numbers in.

1.2 Conventions Used

Symbol / style	Meaning
☐ Checkbox	A task. Do it, then tick it.
Expected result	What you should see on screen after a step. If it does not match, go to Section 14 (Troubleshooting).
Monospace text	Text you type or paste exactly as shown.
Yellow note	A warning or an important limit. Read these before trusting any result.

1.3 Where to Go

If you want to...	Read
Understand the words used	Section 3 (Key Terms)
Open the tool and try it in 10 minutes	Sections 5, 6 and 7
Use your own numbers	Sections 4 and 8
Know how the answers are calculated	Section 9
Compare several options	Section 11
Check that results are believable	Section 12
Share the tool with others through GitHub	Section 15
A one-page summary	Section 16

Time needed: about 30 minutes to complete Sections 4 to 7 with the built-in example.

2. What the Tool Does

A Small Modular Reactor (SMR) is a small nuclear power plant unit that can be built in standard sizes and added one unit at a time. Many countries run older power plants that burn fuel (oil, gas, or coal) to make electricity. Those plants cost money to fuel and release carbon dioxide (CO₂).

The simulator answers one practical question:

If some of a grid's fuel-burning electricity were replaced by SMRs, how much CO₂ and fuel cost would be avoided over the years, and would the SMRs be too large a share of the grid?

Think of it like a household budget simulator: you type in assumptions (income, prices), and it shows what happens if you change one of them. It does not know the real numbers for your situation. It only calculates from what you enter.

2.1 What Goes In and What Comes Out

You enter (inputs)	The tool calculates	You read (outputs)
Current electricity produced by fuel-burning plants; grid size; fuel cost; CO₂ per unit of electricity	Year by year: how much electricity the SMRs can supply, and what that replaces	Cumulative CO ₂ avoided; cumulative fuel cost avoided; SMR share of the grid
SMR unit size, number of units, how fast they are added	How fast SMR output grows	Charts of the electricity mix and grid share over time
Demand growth, years to simulate, a comfort limit for SMR share	Whether SMR share stays under your limit	A SAFE / CAUTION / ABOVE THRESHOLD label

Important: every number that appears when you first open the tool is a demonstration value, not real data for any country. Replace them with your own sourced numbers before drawing conclusions.

3. Key Terms in Plain Language

Term	Plain meaning	Example / unit
Electricity grid	The connected network of power plants and wires that supplies a region or country.	National grid
Thermal power plant	A plant that burns fuel (oil, gas, coal) to make heat, then electricity.	Oil-fired plant
SMR	Small Modular Reactor: a small nuclear plant unit built in standard sizes and added unit by unit.	One unit = 50 to 300 MW here
Megawatt (MW)	A measure of size or power: how much electricity a plant can deliver at one moment. Like the engine size of a car.	100 MW
Gigawatt-hour (GWh)	A measure of energy over time: how much electricity is actually produced. 1 MW running for 1,000 hours = 1 GWh. Like the distance a car actually drove.	1,200 GWh per year
Capacity vs. generation	Capacity (MW) is the maximum a plant could deliver. Generation (GWh) is what it really delivers over a period.	200 MW capacity
Capacity factor	The share of the maximum possible output a plant really achieves over a year, after maintenance and other pauses.	90%
Demand (load)	The electricity people and businesses use. The tool treats it as the total to be supplied.	GWh per year
Load growth	How fast demand rises each year.	3% per year
Emission factor	How much CO ₂ is released for each unit of electricity from a fuel.	0.70 kg CO ₂ per kWh

Term	Plain meaning	Example / unit
Fuel cost	The cost of the fuel needed to make one MWh of electricity (fuel only, no other costs).	140 USD per MWh
Grid share	SMR capacity divided by total grid capacity. A large single source can be risky for a small grid.	33.3%
Threshold	The largest grid share you are willing to accept. You choose it; the tool only compares against it.	15%
Rollout pathway	The order in which SMR units come online: all at once, or spread over years.	Phased
Scenario	One complete set of choices you want to test and compare.	“Phased, 2 units of 100 MW”
Cumulative	The running total over all simulated years.	kilotonnes over 15 years

3.1 Unit Cheat Sheet

Conversion	Value
1 GW	1,000 MW
1 GWh	1,000 MWh = 1,000,000 kWh
1 tonne of CO ₂	1,000 kg
1 kilotonne (kt)	1,000 tonnes
Hours in a year (used by the tool)	8,760

4. Before You Start

4.1 What You Need

- ☐ A computer, tablet, or phone with a modern web browser (Chrome, Edge, Firefox, or Safari).
- ☐ The file SMR_Grid_Integration_Simulator.html on that device.
- ☐ An internet connection the first time you open it (the page loads its charts and fonts from the internet).
- ☐ Somewhere to write down results, such as a notebook or spreadsheet.
- ☐ Optional: your own data (Section 4.2). You can do everything in Sections 5 to 7 without it.

4.2 Information You Need for Your Own Case

Collect these eight numbers, and note where each came from. The tool cannot check them for you.

Information	Unit	Where to look
Electricity produced per year by the fuel-burning plants you want to replace	GWh/yr	National utility or ministry annual reports; statistics from international bodies such as IEA, IRENA, the World Bank, or the U.S. EIA
Total capacity of all plants on the grid	MW	Same sources as above

Information	Unit	Where to look
CO ₂ released per unit of electricity from that fuel	kg CO ₂ /kWh	Plant or utility records, or published default emission factors (for example from the IPCC)
Fuel cost per unit of electricity	USD/MWh	Fuel price × fuel used per MWh, from utility or plant data
SMR unit size you want to test	MW	Public SMR design data sheets from reactor developers or the IAEA
SMR capacity factor	%	Published operating statistics or design data
Expected yearly growth in electricity demand	%/yr	National energy plans or forecasts
Your comfort limit for SMR share of the grid	%	A grid code or planning rule if one exists; otherwise your own documented assumption

Good practice: for every number write down the source, the date you looked, and the year the number refers to. Results are only as trustworthy as the inputs.

5. Opening the Tool

- Find the file `SMR_Grid_Integration_Simulator.html` in your files or downloads folder.
- Double-click it. If it opens as text or in an editor, instead right-click it and choose Open With, then pick your web browser.
Expected result: *Your browser opens a dark-blue page titled “SMR Grid Integration Simulator”.*
- Wait a few seconds for the page to finish loading.
Expected result: *A yellow banner reads “Every number below is a placeholder.” Four number cards, a coloured bar, and three charts appear on the right. A control panel is on the left.*
- Look at the top-right corner of the page.
Expected result: *It shows the version (v1.0, first version) and the names of the authors.*

If the charts stay empty: you are probably offline. Connect to the internet and reload the page (Section 14).

Opening the file from a terminal, or putting it online, is covered in Section 15.

6. A Tour of the Screen

Area	What it is for
Yellow banner (top)	A permanent reminder that the starting numbers are placeholders.
Control panel (left; above the results on a phone)	Every input. This is the only place you type or move sliders. There is no Run button: the results update the moment you change anything.
Four number cards	The headline results: SMR capacity, grid share, CO ₂ avoided, fuel cost avoided.
Grid capacity bar	A picture of the final year: striped part = other capacity, teal part = SMR, amber line = your threshold.
Three charts	How things change year by year: electricity mix, grid share against your threshold, and running totals of CO ₂ and cost avoided.

Area	What it is for
Scenario comparison (bottom)	A table and chart where you keep up to four saved scenarios side by side.
Methodology (bottom, click to open)	The exact formulas the tool uses.

7. Quick Start: Run the Built-In Example

This 10-minute exercise uses the numbers already in the tool. It teaches the controls and lets you check that everything works. Your results should match those below exactly.

7.1 Check the Starting Values

- ☐ Baseline annual generation = 1200 | Grid installed capacity = 600 | Emission factor = 0.70 | Fuel cost = 140
- ☐ Unit size = 100 MW | Number of units = 2 | Rollout pathway = Phased multi-unit | Capacity factor = 90%
- ☐ Load growth = 3.0%/yr | Simulation horizon = 15 yrs | Safe grid-share threshold = 15%

If any value differs, click Reset to defaults at the bottom of the control panel.

7.2 Read the Result Cards

Card	Expected result
Total SMR Capacity	200 MW ("2 × 100 MW units (phased)")
Grid Share (final year)	33.3%, with the label ABOVE THRESHOLD
CO ₂ Avoided (cumulative)	15,650.8 kt ("1,103,760 t/yr in final year")
Fuel Cost Avoided (cumulative)	\$3,130.2M ("-\$220.75M/yr in final year")

7.3 Change One Thing at a Time

Do these in order. After each one, compare what you see with the expected result.

- Change Rollout pathway to "Single unit, one-time".
Expected result: CO₂ = 16,252.2 kt; cost = \$3,250.4M; grid share still 33.3%. All capacity is available from the start, so more is avoided, but the final grid share is the same.
- Now move the Number of units slider to 1.
Expected result: Capacity = 100 MW; grid share = 16.7% with the label CAUTION; CO₂ = 8,830.1 kt; cost = \$1,766M.
- Set Rollout pathway to "Phased multi-unit" and Number of units to 3.
Expected result: Capacity = 300 MW; grid share = 50%; CO₂ = 16,330.3 kt; cost = \$3,266.1M. Compare with the very first result: adding a third unit added little, because SMR output cannot exceed demand (see Section 9).
- Click Reset to defaults.
Expected result: Every value and result returns to Section 7.2.
- Change Fuel cost to 200.
Expected result: Cost avoided rises to \$4,471.7M; CO₂ stays 15,650.8 kt. Fuel cost affects money only, not CO₂.

Result: if all five checks matched, the tool is working correctly on your device and you understand the basic controls.

8. Entering Your Own Data

Work down the control panel from top to bottom. The four boxes at the top accept any number above zero. The dropdowns and sliders below them only allow the ranges shown.

Field	Plain meaning	Unit	Allowed values	Where the number comes from
Baseline annual generation	Electricity now made in a year by the fuel-burning plants. The tool treats it as demand in Year 0 and assumes fuel plants supply whatever the SMRs do not.	GWh/yr	Above 0 (starts at 1200)	Utility or ministry annual statistics
Grid installed capacity	Combined size of all power plants on the grid. Used only to work out grid share.	MW	Above 0 (starts at 600)	Utility or national statistics
Emission factor	CO ₂ released for each kWh made from the fuel the SMRs replace.	kg CO ₂ /kWh	Above 0 (starts at 0.70)	Plant records or published default factors
Fuel cost	Fuel cost of each MWh made by the fuel plants. Do not enter a price per barrel or per tonne.	USD/MWh	Above 0 (starts at 140)	Fuel price × fuel needed per MWh
Unit size	Size of one SMR unit.	MW	50, 100, 150, 200 or 300	SMR design data sheets
Number of units	How many units are built in this scenario.	count	1 to 4	Your scenario choice
Rollout pathway	How the units are added over time (table below).	choice	Single / Phased / Distributed	Your scenario choice
Capacity factor	Share of maximum output the SMRs really deliver over a year.	%	60 to 97	Published operating data
Load growth rate	Yearly rise in demand.	%/yr	0 to 8, steps of 0.5	National energy plans
Simulation horizon	How many years to simulate. A horizon of N covers Year 0 to Year N, which is N+1 yearly values.	years	5 to 20	Your study period
Safe grid-share threshold	The largest SMR share of grid capacity you accept.	%	5 to 30	Grid code or your documented assumption

8.1 The Three Rollout Pathways

Pathway	Exactly what the tool does
Single unit, one-time	All units are online from Year 0.
Phased multi-unit	The first unit is online in Year 0. One more unit is added every 2 years until all are online.

Pathway	Exactly what the tool does
Distributed smaller units	The first unit is online in Year 0. One more unit is added every year until all are online. This is a timing simplification only; it does not model different geographic sites.

8.2 Data-Entry Checklist

- ☐ Every box has a number. An empty box makes the results meaningless.
- ☐ Each number uses the unit shown beside the box (GWh, MW, kg CO₂/kWh, USD/MWh).
- ☐ All numbers refer to the same year and the same group of plants.
- ☐ You wrote down the source of each number.
- ☐ You change only one input at a time when comparing options.

No real data? you can still explore. Use clearly labelled assumptions and describe any results as “illustrative”, never as forecasts.

9. How the Results Are Produced

The tool repeats the same seven steps for every year from Year 0 to your chosen horizon. Nothing is hidden: the same formulas are shown in the Methodology panel at the bottom of the tool.

Step	What is worked out	How
1	Demand	Baseline generation × (1 + load growth) raised to the power of the year number.
2	SMR capacity online	From the rollout pathway (Section 8.1) and the number and size of units.
3	Most the SMRs could supply (GWh)	Capacity (MW) × 8,760 hours × capacity factor ÷ 1,000.
4	What the SMRs actually supply (GWh)	The smaller of step 3 and demand. Fuel-burning plants supply the rest.
5	CO ₂ avoided (tonnes)	SMR supply (GWh) × 1,000 × emission factor.
6	Fuel cost avoided (USD)	SMR supply (GWh) × 1,000 × fuel cost.
7	Grid share (%)	SMR capacity online ÷ grid installed capacity × 100.

The cumulative cards add up the yearly values of steps 5 and 6 across all years (Year 0 to Year N).

9.1 Worked Example You Can Check by Hand

Using the starting values (Section 7.1):

Quantity	Year 0	Year 15
Demand (GWh)	1,200	$1,200 \times 1.03^{15} = 1,869.6$
SMR units online (phased)	1 unit = 100 MW	2 units = 200 MW
Most the SMRs could supply (GWh)	$100 \times 8,760 \times 0.90 \div 1,000 = 788.4$	$200 \times 8,760 \times 0.90 \div 1,000 = 1,576.8$
SMR supply (GWh)	788.4 (below demand)	1,576.8 (below demand)

Quantity	Year 0	Year 15
CO ₂ avoided	$788.4 \times 1,000 \times 0.70 = 551,880 \text{ t}$	$1,576.8 \times 1,000 \times 0.70 = 1,103,760 \text{ t}$
Fuel cost avoided	$788.4 \times 1,000 \times 140 = \110.38M	$1,576.8 \times 1,000 \times 140 = \220.75M
Grid share	$100 \div 600 = 16.7\%$	$200 \div 600 = 33.3\%$

The Year 15 CO₂ and cost match the “final year” lines under the result cards in Section 7.2.

9.2 Why a Bigger SMR Fleet Does Not Always Help Much More

In Year 15, three 100 MW units could supply up to $300 \times 8,760 \times 0.90 \div 1,000 = 2,365.2$ GWh, but demand is only 1,869.6 GWh. Step 4 caps the SMR supply at demand, so the third unit only adds 292.8 GWh compared with two units. Once SMRs can meet all demand, extra capacity brings no extra benefit in this tool.

10. Reading the Results

10.1 The Four Number Cards

Card	What it tells you
Total SMR Capacity	Unit size $\times$ number of units. Use it to check that you set the scenario you meant.
Grid Share (final year)	SMR capacity as a percentage of grid capacity at the end of the horizon, with a label (below).
CO ₂ Avoided (cumulative)	Total CO ₂ avoided over all years, in kilotonnes (kt). The line beneath gives tonnes for the final year.
Fuel Cost Avoided (cumulative)	Total fuel cost avoided over all years, in millions of USD. The line beneath gives the final year.

10.2 The Grid Share Label

Label	When it appears
SAFE	Final-year grid share is at or below your threshold.
CAUTION	Final-year grid share is above your threshold but no more than 1.5 times it.
ABOVE THRESHOLD	Final-year grid share is more than 1.5 times your threshold.

Example: with a 15% threshold, 16.7% shows CAUTION and 33.3% shows ABOVE THRESHOLD. The label looks only at the final year. The Grid Share chart shows every year.

10.3 The Bar and the Three Charts

- Grid capacity bar: teal = SMR share, striped = all other capacity, amber line = your threshold. If teal passes the amber line, the label is not SAFE.
- Generation Mix Over Time: stacked areas of fuel-plant and SMR electricity (GWh per year). The total height equals demand. SMR should never rise above the total.
- Grid Share vs. Safe Threshold: SMR share each year against a dashed amber line.

- Cumulative CO₂ and Fuel Cost Avoided: bars show cumulative CO₂ (tonnes, left axis); the line shows cumulative fuel cost (USD, right axis). Hover over a point to read its value.

11. Testing and Comparing Scenarios

11.1 A Reliable Method

- ☐ Set up your baseline case and click Save this scenario.
- ☐ Change exactly one input (for example the rollout pathway).
- ☐ Click Save this scenario again.
- ☐ Write down every setting you used for each saved scenario (see 11.2).
- ☐ Compare the rows in the table and chart at the bottom of the page.

11.2 How Saving Works

- Up to 4 scenarios can be kept. Click the ✕ beside a row to delete it and free a slot.
- The name is created automatically from the pathway, number of units and unit size only (for example “Phased · 2×100MW”). Two scenarios that differ only in fuel cost, emission factor, capacity factor or growth will have the same name, so keep notes.
- The comparison table stores results (capacity, grid share, CO₂, cost). It does not display the other settings.
- Scenarios are kept in this browser on this computer. Private or incognito windows, clearing browsing data, or another browser or device will not keep them. Write down important results.
- Reset to defaults resets the inputs only; it does not delete saved scenarios.

11.3 Practice Set

Starting from the defaults (Section 7.1), save these four scenarios in order. Change only what is listed.

Slot	Settings	Capacity	Grid share	CO ₂ (cumulative)	Fuel cost (cumulative)
1	Phased, 2 units of 100 MW (the defaults)	200 MW	33.3%	15,650.8 kt	\$3,130.2M
2	Single unit, one-time, 2 units	200 MW	33.3%	16,252.2 kt	\$3,250.4M
3	Single unit, one-time, 1 unit	100 MW	16.7%	8,830.1 kt	\$1,766M
4	Phased, 3 units	300 MW	50%	16,330.3 kt	\$3,266.1M

What the practice set shows:

- Slots 1 and 2: the same capacity avoids about 4% more CO₂ when both units start in Year 0, because the second unit also contributes in Years 0 and 1.
- Slots 1 and 3: halving the capacity cuts the cumulative CO₂ avoided by about 44% (15,650.8 kt to 8,830.1 kt) and halves the grid share.
- Slots 1 and 4: 50% more capacity avoids only about 4% more CO₂, because SMR output is capped by demand (Section 9.2) while the grid share rises from 33.3% to 50%.

12. Are the Results Believable? A Checklist

Run through this list before you show or quote any result.

- ☐ Units: every input uses the unit shown beside its box (Section 8).
- ☐ Nothing empty: no input box is blank.
- ☐ Mix chart: the SMR area never rises above the total height of the chart (total = demand).
- ☐ Grid share: it is between 0% and 100%. Above 100% means SMR capacity is larger than the grid you entered.
- ☐ Hand check: recalculate one year using the seven steps in Section 9 and confirm it matches the tool.
- ☐ Direction check: the changes below move the results the way the table says.
- ☐ Sources: every input has a written source, date, and year.
- ☐ Wording: results are described as illustrations from stated assumptions, not predictions.

If you increase...	This should change	This should NOT change
Fuel cost	Fuel cost avoided rises	CO ₂ avoided; grid share
Emission factor	CO ₂ avoided rises	Fuel cost avoided; grid share
Grid installed capacity	Grid share falls	CO ₂ avoided; fuel cost avoided
Capacity factor	CO ₂ and cost avoided rise, unless SMRs already meet all demand	Grid share
Unit size or number of units	Capacity and grid share rise; CO ₂ and cost avoided rise, unless SMRs already meet all demand	—
Load growth rate	Demand rises; CO ₂ and cost avoided stay the same or rise	Grid share (the tool keeps grid capacity fixed)

13. What the Tool Does Not Do

Read this before using any result in a report: this is a first-pass screening calculator, not a full engineering or economic study.

- It does not test grid stability. It does not model frequency response, sudden plant outages, or power flow through the wires. Grid share is only a simple warning indicator.
- Cost avoided is gross fuel cost avoided. It does not subtract the cost of building, operating, fuelling, or decommissioning the SMRs, so it is not the net saving from the project.
- Emissions from building and fuelling SMRs are counted as zero.
- Fuel cost, emission factor, and capacity factor stay constant in every year.
- Grid installed capacity stays fixed while demand grows.
- The tool assumes fuel-burning plants supply all demand that the SMRs do not.
- Rollout pathways are simplified timings. “Distributed” does not model real locations.
- It uses one generic SMR unit size at a time; it does not model any specific reactor design.
- Nothing here is a forecast, recommendation, or a substitute for a professional study.

14. Troubleshooting

Problem	What to do
Charts are empty or missing	Connect to the internet and reload the page. The chart library and fonts load from the internet on first use.
The file opens as text or code	Close it. Right-click the file, choose Open With, and pick a web browser (Chrome, Edge, Firefox, or Safari).
Cards show NaN or strange values	An input box is empty or contains text. Enter a number in every box, or click Reset to defaults.
Grid share is above 100%	SMR capacity is larger than the grid capacity you entered. Check that both are in MW and describe the same grid.
A result did not change after I changed an input	Either the input does not affect that result (see the table in Section 12), or SMR supply is already capped at demand (Section 9.2).
Saved scenarios have disappeared	They are stored per browser and per computer. A private window, cleared browsing data, or a different browser or device will not show them. Save important results in your own notes.
Two saved scenarios have the same name	Names use only the pathway, number of units, and unit size. Keep your own notes of the other settings.
I want to start over	Click Reset to defaults. This resets inputs only; use the ✕ beside a row to delete saved scenarios.
The page looks cramped on a phone	On a small screen the control panel appears above the results. Scroll down to see the cards and charts.

15. Sharing the Tool Through GitHub

GitHub is a free website for storing and sharing files. Uploading the tool there lets other people download it or open it as a web page. Replace <org> with the GitHub account or organisation name and <repo> with the repository name.

15.1 Upload the File

Option A: no commands needed.

1. Sign in to github.com and open your repository, or create a new one.
2. Click Add file, then Upload files.
3. Drag in SMR_Grid_Integration_Simulator.html (and this guide, if you want).
4. Type a short message, for example “Add SMR simulator v1.0”, and click Commit changes.

Expected result: *The file appears in the repository’s file list.*

Option B: using Git in a terminal (Git must be installed).

```
git clone https://github.com/<org>/<repo>.git
cd <repo>
cp /path/to/SMR_Grid_Integration_Simulator.html .
git add SMR_Grid_Integration_Simulator.html
git commit -m "Add SMR simulator v1.0"
git push origin main
```

Public or private? a public repository can be seen by anyone. Choose deliberately before uploading, especially once your own data is in use.

15.2 Open the File From a Terminal

After downloading or cloning, run the command for your system from the folder that contains the file. Double-clicking the file works on every system too.

```
macOS:          open SMR_Grid_Integration_Simulator.html
Windows (cmd):   start SMR_Grid_Integration_Simulator.html
Windows (PowerShell): Invoke-Item SMR_Grid_Integration_Simulator.html
Linux:           xdg-open SMR_Grid_Integration_Simulator.html
```

Expected result: your default browser opens the dashboard (Section 5).

15.3 Share It as a Web Page (GitHub Pages)

Why this is needed: clicking the file inside GitHub shows its source code, not the working dashboard. To see the dashboard, download the file or publish it with GitHub Pages.

1. In the repository, open Settings, then Pages.
2. Under Build and deployment, set Source to Deploy from a branch.
3. Set Branch to main and the folder to / (root), then click Save.
4. Wait one to two minutes, then open the address GitHub shows on that page.

Expected result: *The address has this form:*

```
https://<org>.github.io/<repo>/SMR_Grid_Integration_Simulator.html
```

Quick alternative without setup: a third-party preview service, htmlpreview.github.io, can display a public file. It is slower and best used only for a quick review.

```
https://htmlpreview.github.io/?https://github.com/<org>/<repo>/blob/main/SMR_Grid_Integration_Simulator.html
```

15.4 Optional: Open It Through a Local Web Server

If your browser blocks anything when opening the file directly, run this in the folder containing the file (Python 3 must be installed), then open the address in a browser.

```
python3 -m http.server 8000
http://localhost:8000/SMR_Grid_Integration_Simulator.html
```

Press Ctrl+C in the terminal to stop the server.

16. Quick Reference

16.1 The Ten-Step Routine

1. Open the tool in a browser and read the yellow banner.

2. Click Reset to defaults.
3. Enter the four baseline numbers: generation, grid capacity, emission factor, fuel cost.
4. Choose unit size, number of units, rollout pathway, and capacity factor.
5. Set load growth, horizon, and your grid-share threshold.
6. Read the four cards, the bar, and the three charts.
7. Run the checks in Section 12.
8. Click Save this scenario and note the settings.
9. Change one input, save again, and compare.
10. Report results as illustrations, with sources for every input and the limits in Section 13.

16.2 Formulas (Each Year, Year 0 to Year N)

Quantity	Formula
Demand (GWh)	$\text{Baseline} \times (1 + \text{growth})^{\text{year}}$
Most SMRs could supply (GWh)	$\text{Capacity (MW)} \times 8,760 \times \text{capacity factor} \div 1,000$
SMR supply (GWh)	Smaller of the line above and demand
CO ₂ avoided (tonnes)	$\text{SMR supply (GWh)} \times 1,000 \times \text{emission factor (kg/kWh)}$
Fuel cost avoided (USD)	$\text{SMR supply (GWh)} \times 1,000 \times \text{fuel cost (USD/MWh)}$
Grid share (%)	$\text{SMR capacity online} \div \text{grid capacity} \times 100$

16.3 Label Rules

Label	Rule (final year)
SAFE	$\text{Share} \leq \text{threshold}$
CAUTION	$\text{Threshold} < \text{share} \leq 1.5 \times \text{threshold}$
ABOVE THRESHOLD	$\text{Share} > 1.5 \times \text{threshold}$